

Updated code:

```
.data
result: .word 0    # keep only final result in memory
```

```
.text
.globl main
main:
```

```
# a = 3 (keep in $t0)
li $t0, 3
```

```
# b = a + 2 (keep in $t1)
addi $t1, $t0, 2
```

```
# result = a + b + 5
add $t2, $t0, $t1 # t2 = a + b
addi $t2, $t2, 5  # t2 = t2 + 5
sw $t2, result
```

```
# print result
lw $a0, result
li $v0, 1    # syscall: print_int
syscall
```

```
# exit
li $v0, 10
syscall
```

Result below:

QtSpim

File Simulator Registers Text Segment Data Segment Window Help

FP Regs Int Regs [16] Data Text

Int Regs [16]

PC = 400050
EPC = 0
Cause = 0
BadVAddr = 0
Status = 3000ff10

HI = 0
LO = 0

R0 [r0] = 0
R1 [at] = 10010000
R2 [v0] = a
R3 [v1] = 0
R4 [a0] = d
R5 [a1] = 7ffff734
R6 [a2] = 7ffff73c
R7 [a3] = 0
R8 [t0] = 3
R9 [t1] = 5
R10 [t2] = d
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 0
R17 [s1] = 0
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
R22 [s6] = 0

User data segment [10000000]..[10040000]

[10000000]..[1000ffff] 00000000
[10010000] 0000000d 00000000 00000000 00000000
[10010010]..[1003ffff] 00000000

User Stack [7ffff730]..[80000000]

[7ffff730] 00000001 7ffff7f8 00000000 7fffffe1
[7ffff740] 7fffffba 7fffff93 7fffff68 7fffff37
[7ffff750] 7fffffb7 7fffffca 7fffff9d 7fffff6a
[7ffff760] 7fffff63 7ffffe4d 7fffff37 7fffff0a
[7ffff770] 7ffffe00 7ffffdee 7ffffcf3 7fffff6a
[7ffff780] 7ffffd8c 7ffffd6d 7ffffcf3 7fffff6a
[7ffff790] 7fffface 7ffffabe 7ffffa46 7ffffa2e
[7ffff7a0] 7ffffa46 7ffffa2e 7ffff9cc 7ffff9ae
[7ffff7b0] 7ffff9cc 7ffff9ae 7ffff91e 7ffff90a
[7ffff7c0] 7ffff91e 7ffff90a 7ffff8be 7ffff898
[7ffff7d0] 7ffff8be 7ffff898 7ffff85c 7ffff841
[7ffff7e0] 7ffff85c 7ffff841 00000000 2d353034
[7ffff7f0] 00000000 2d353034 70676a2f 442f3231
[7ffff800] 70676a2f 442f3231 62616c2f 612e3031
[7ffff810] 62616c2f 612e3031 4c42414e 59535f45
[7ffff820] 4c42414e 59535f45 69646e69 3a433d72
[7ffff830] 69646e69 3a433d72 45535500 4f525052
[7ffff840] 45535500 4f525052 72657355 676a5c73
[7ffff850] 72657355 676a5c73 454d414e 70676a3d
[7ffff860] 454d414e 70676a3d 55003231 44524553
[7ffff870] 49414d4f 4f525f4e 4e494d41 4f525047
[7ffff880] 454c4946 50484a3d 45535500 4d4f4452
[7ffff890] 3d4e4941 0050484a 3d504d54 555c3a43
[7ffff8a0] 73726573 70676a5c 415c3231 61447070
[7ffff8b0] 4c5c6174 6c61636f 6d65545c 45540070
ta\Local\Temp.TE

Console

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